SUMAN ELECTRONICS

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MODBUS DATA FORMAT (Document Version: 01/01)

Communication Parameters:

1. Register Length : 26
2. Word Length (Data Length) : 8 Bit
3. Parity : None [Disabled]
4. Stop Bits : 2
6. Baud Rate : 9600 bits / second
6. Transmission Mode : RTU
7. Slave Response Delay : 750 milli seconds, minimum

Holding Reg Number	Description	Properties	Details / Range
1	PC SET TEMPERATURE REGISTER	READ / WRITE	19 to 32 Degree Celcius
2	PC OPERATING MODE REGISTER	READ / WRITE	Details Mentioned Below
3	RESET COMMAND FROM PC	READ / WRITE	1 = Reset Fault, Else Donot Reset Fault
4	CONTROL OPTION REGISTER	READ / WRITE	0 = Control From PC, Else from Handset
5	HANDSET CONTROL TIME	READ / WRITE	1 to 65535 Minutes
6	NUMBER OF COMPRESSORS IN THE SYSTEM	READ ONLY	Details Mentioned Below
7	MODEL TYPE	READ ONLY	Details Mentioned Below
8	CONTROLLER SET TEMPERATURE REGISTER	READ ONLY	19 to 32 Degree Celcius
9	CONTROLLER OPERATING MODE REGISTER	READ ONLY	Details Mentioned Below
10	FAULT RESETTING CONDITION ACTIVE	READ ONLY	1 = Reset Fault Active, Else Deactive
11	RETURN AIR TEMPERATURE	READ ONLY	
12	COMPRESSOR 1 RUN HOURS	READ ONLY	
13	COMPRESSOR 2 RUN HOURS	READ ONLY	
14	BLOWER FAN RUN HOURS	READ ONLY	
15	COMPRESSOR STATUS REGISTER	READ ONLY	Details Mentioned Below
16	BLOWER FAN STATUS REGISTER	READ ONLY	Details Mentioned Below
17	ALARM STATUS REGISTER	READ ONLY	Details Mentioned Below
18	SYSTEM FAULT REGISTER	READ ONLY	Details Mentioned Below
19	ANALOG SENSOR FAULT REGISTER	READ ONLY	Details Mentioned Below
20	COMPRESSOR 1 FAULT REGISTER	READ ONLY	Details Mentioned Below
21	COMPRESSOR 2 FAULT REGISTER	READ ONLY	Details Mentioned Below
22	SUPPLY AIR TEMPERATURE	READ ONLY	

23	HUMIDITY 1	READ ONLY	
24	HUMIDITY 1	READ ONLY	
25	COMPRESSOR 1 CURRENT	READ ONLY	
26	COMPRESSOR 2 CURRENT	READ ONLY	

BIT DETAILS:

The data that is transmitted from Slave controller to Master contains following information:

1. Number of Compressors

If number of compressors is to be changed from PC, PC should write this register in following format:

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

D0	Description
0	1 Compressor System
1	2 Compressor System

2. Model Type

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

If model type is to be changed from PC, PC should write this register in following format:

D0	Description	Bit Applicable To
0	Bluestar Machine (Common HPLP)	IDU
1	Carrier Machine (Separate HPLP)	

D1	Description	Bit Applicable To
0	SPPR Sensed from machine's 3 Phase Monitoring Module	IDU
1	SPPR Sensed from External SPPR input Connected to HP1 Input	

Note: If value of Holding Register 1 or Holding Register 2 is modified, it is assumed that new machine is configured, and so, run hours of fan and compressors is reset to 0.

D2	Description	Bit Applicable To
0	System Without ODU	IDU
1	System With ODU	

3. Compressor Status

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	COMPRESSOR 1 OFF	IDU and Package
		1	COMPRESSOR 1 ON	
2	D1	0	COMPRESSOR 2 OFF	
		1	COMPRESSOR 2 ON	
3	D2	0	COMPRESSOR 3 OFF	
		1	COMPRESSOR 3 ON	

4. Blower Fan Status

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	BLOWER FAN OFF	Package
		1	BLOWER FAN ON	

In case of IDU machines, following is bit allocation to indicate actual fan speed:

Sr. No.	D2	D1	Description
1	0	0	Blower Fan Running in Low Speed
2	0	1	Blower Fan Running in Medium Speed
3	1	0	Blower Fan Running in High Speed
4	1	1	Future Use

Register number 22 is setting register while register number 12 is actual status of blower fan.

5. Alarm Status

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	ALARM OFF	All Models
		1	ALARM ON	

This register indicates whether system is having any alarm or no. If any alarm is there, this register will have value 1, else it would have value 0.

6. System Fault

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	SPPR FAULT CLEARED	IDU
		1	SPPR FAULT EXIST	
2	D1	0	BF SPPR FAULT CLEARED	IDU
		1	BF SPPR FAULT EXIST	
3	D2	0	ODU-1 COMM ERROR CLEARED	IDU
		1	ODU-1 COMM ERROR EXIST	
4	D3	0	ODU-2 COMM ERROR CLEARED	IDU
		1	ODU-2 COMM ERROR EXIST	

Note: Bits D1, D2 and D3 are applicable if System with ODU is configured.

7. Analog Sensor Fault Register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	RAT OPEN FAULT CLEARED	IDU
		1	RAT OPEN FAULT EXIST	
2	D1	0	SAT OPEN FAULT CLEARED	IDU

		1	SAT OPEN FAULT EXIST	
3	D2	0	HUMIDITY 1 OPEN FAULT CLEARED	IDU
		1	HUMIDITY 1 OPEN FAULT EXIST	
3	D3	0	HUMIDITY 2 OPEN FAULT CLEARED	IDU
		1	HUMIDITY 2 OPEN FAULT EXIST	

8. Blower Fan Fault

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	BLOWER FAN FAULT EXIST	Package
		1	BLOWER FAN FAULT CLEARED	

9. Compressor 1 Fault

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	ODU 1 SPPR CLEARED	IDU
		1	ODU 1 SPPR CLEARED	
2	D1	0	HP /LP 1 FAULT CLEARED	AIRCOOL
		1	HP/LP 1 FAULT EXIST	
3	D2	0	HP 1 FAULT CLEARED	SAPERATE HP & LP
		1	HP 1 FAULT EXIST	
4	D3	0	LP 1 FAULT CLEARED	SAPERATE HP & LP
		1	LP 1 FAULT EXIST	
5	D4	0	ODU 1 OVERLOAD CLEARED	IDU
		1	ODU 1 OVERLOAD EXIST	
6	D5	0	ODU 1 UNDERLOAD CLEARED	IDU
		1	ODU 1 UNDERLOAD EXIST	

D1 bit would be used for Bluestar machines, D2 and D3 bits would be used for Carrier machines.

10. Compressor 2 Fault

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	ODU 2 SPPR CLEARED	IDU
		1	ODU 2 SPPR CLEARED	
2	D1	0	HP /LP 2 FAULT CLEARED	AIRCOOL
		1	HP/LP 2 FAULT EXIST	
3	D2	0	HP 2 FAULT CLEARED	SAPERATE HP & LP
		1	HP 2 FAULT EXIST	
4	D3	0	LP 1 FAULT CLEARED	SAPERATE HP & LP
		1	LP 2 FAULT EXIST	
5	D4	0	ODU 2 OVERLOAD CLEARED	IDU
		1	ODU 2 OVERLOAD EXIST	
6	D5	0	ODU 2 UNDERLOAD CLEARED	IDU
		1	ODU 2 UNDERLOAD EXIST	

D1 bit would be used for Bluestar machines, D2 and D3 bits would be used for Carrier machines.

11. Operating Mode

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Sr. No.	Bit Name	Bit Value	Description	Bit Applicable To
1	D0	0	OFF MODE	IDU
		1	ON MODE	
2	D5	0	FAN MODE	
		1	COOL MODE	

Sr. No.	D2	D1	Description
1	0	0	Fan Speed: Low
2	0	1	Fan Speed: Medium
3	1	0	Fan Speed: High
4	1	1	Future Use

12. Fault Resetting Condition Active:

If, any fault exists, and, in case fault could be reseted on controller side, controller will write this register as 1, else this register would be 0.

Note: PC should check value of this register and should activate fault reset procedure from PC side if value of this register is 1.

13. Reset Command from PC:

For resetting any fault from PC, this register should be written as 1. Else it should be written as 0.